

Miscellaneous

Yao Zhang

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1 Recreation

1.1 This Is Us

1.2 Vocabulary

1.3 Conversation

1. 1700
2. 1500
3. 1300

1.4 Driving Experience

座椅调节顺序: 上下 → 前后 → 靠背. 方向盘调节顺序: 前后 → 上下. 后视镜调节顺序: 上下 → 左右.

- | | | |
|---------------|----------------|----------------|
| 1. 坐姿后视镜调节 | 11. 车头车尾距障碍物距离 | 21. 雨天开车 |
| 2. 自动挡正确起步和停车 | 12. 倒车入库 | 22. 雾天开车 |
| 3. 方向盘正确打法 | 13. 开车出库 | 23. 跑高速流程 |
| 4. 踩刹车技巧 | 14. 侧方位停车 | 24. 跑高速注意事项 |
| 5. 新手开车注意事项 | 15. 窄路侧方位停车 | 25. 雨天跑高速 |
| 6. 开车安全口诀 | 16. 靠边停车 | 26. 雾天跑高速 |
| 7. 提升车感 | 17. 掉头 | 27. 雪后跑高速 |
| 8. 灯光正确操作 | 18. 窄路判断车距 | 28. 夏天开车注意事项 |
| 9. 新手开车错误习惯 | 19. 农村开车技巧 | 29. 冬天开车正确使用空调 |
| 10. 停车技巧 | 20. 市区开车技巧 | 30. 手机导航 |

1.5 The Big Bang Theory

1.6 Popular Science Series

1. 杨振宁: 美与物理学
2. 杨振宁: 20 世纪数学与物理的分与合
3. 丘成桐: 广义相对论与数学
4. 丘成桐: 时空几何学与广义相对论

5. 高涌泉: 比相对论更奇怪的量子力学
 6. 高涌泉: 爱因斯坦如何发现狭义相对论
 7. 高涌泉: 20 世纪物理大师理查德·费曼其人其事
 8. 高涌泉: 为什么非弯曲时空不可? 爱因斯坦的洞见
 9. 高涌泉: 量子力学究竟怪在哪里? 海森堡在 100 年前开启的学问
 10. 张庆瑞: 量子加 AI, 新文明诞生
 11. 张庆瑞: 量子纠缠, 由哲学到数学, 再经科学到科技
 12. 李琳山: 信号与人生 2019 2020 2021 2022 2023 2025
 13. 张海潮: 爱因斯坦的科学 with 生平
 14. Chad Davies: Maxwell's Equations
 15. 邵锦昌: 基础科学研究方法: 跨领域研究实例
 16. 陈义裕: 玄妙时空理论的绚丽展现
 17. 耿朝强: 宇宙学与粒子物理前瞻问题
 18. 沈成平: 从夸克到上帝粒子用粒子物理打开世界真相
 19. 余舜德: 为什么人类学需要脑神经科学
 20. WSA2025: The Future of Stroke Treatment: Insights from AI and Emerging Technologies
 21. M. Poluekto & A. Polar: Efficient training of Kolmogorov-Arnold Networks
 22. 时间管理 A, B
 23. 宇宙生命科技
 24. 访谈: 原生家庭
-
- 1.7 The Story of the Stone
 - 1.8 New Concept English Series
 - 1.9 Romance of the Three Kingdoms

1. 宴桃园豪杰三结义, 斩黄巾英雄首立功
2. 张翼德怒鞭督邮, 何国舅谋诛宦竖
3. 议温明董卓叱丁原, 馈金珠李肃说吕布
4. 废汉帝陈留为皇, 谋董贼孟德献刀
5. 发矫诏诸镇应曹公, 破关兵三英战吕布
6. 焚金阙董卓行凶, 匿玉玺孙坚背约

2 Life Sciences

2.1 Cardiology

2.2 Brain Imaging

2.3 Clinical Neurology

2.4 Biomedical Engineering

2.5 Medical Image Computing

2.6 Maternal and Child Health

2.6.1 Job Specifications

- | | |
|-----------|-----------|
| 1. 岗位认知 A | 3. 岗位要求 A |
| 2. 岗位认知 B | 4. 岗位要求 B |

2.6.2 Infant Care

- | | |
|--------------------|--------------------|
| 1. 新生儿的定义及类别 A | 13. 婴儿非疾病的多种异常情况 D |
| 2. 新生儿的定义及类别 B | 14. 婴儿非疾病的多种异常情况 E |
| 3. 新生儿的生长指标及生理特点 A | 15. 满月婴儿的发育特点及指标 |
| 4. 新生儿的生长指标及生理特点 B | 16. 婴儿家庭护理准备 |
| 5. 新生儿的能力发育 | 17. 婴儿日常护理 A |
| 6. 婴儿的生理反射和几种状态 | 18. 婴儿日常护理 B |
| 7. 婴儿的营养需求 | 19. 婴儿日常护理 C |
| 8. 婴儿的喂养方式 A | 20. 婴儿日常护理 D |
| 9. 婴儿的喂养方式 B | 21. 婴儿日常护理 E |
| 10. 婴儿非疾病的多种异常情况 A | 22. 婴儿日常护理 F |
| 11. 婴儿非疾病的多种异常情况 B | 23. 婴儿日常护理 G |
| 12. 婴儿非疾病的多种异常情况 C | 24. 婴儿日常护理 H |

- 25. 婴儿日常护理 I
- 26. 新生儿常见问题及紧急情况处理 A
- 27. 新生儿常见问题及紧急情况处理 B
- 28. 婴儿早教和智力开发 A
- 29. 婴儿早教和智力开发 B
- 30. 婴儿早教和智力开发 C
- 31. 婴儿日常护理: 游泳
- 32. 婴儿日常护理: 疾病的预防与观察
- 33. 婴儿常见疾病的观察和护理 A
- 34. 婴儿常见疾病的观察和护理 B
- 35. 婴儿常见疾病的观察和护理 C
- 36. 婴儿常见疾病的观察和护理 D
- 37. 婴儿常见疾病的观察和护理 E
- 38. 婴儿常见疾病的观察和护理 F
- 39. 婴儿出行安全
- 40. 早产儿
- 41. 先天性异常儿 A
- 42. 先天性异常儿 B
- 43. 婴儿护理实践操作

2.6.3 Maternal Care

- 1. 产褥期产妇身体变化特点 A
- 2. 产褥期产妇身体变化特点 B
- 3. 产褥期的临床表现 A
- 4. 产褥期的临床表现 B
- 5. 产褥期护理的重要性 A
- 6. 产褥期护理的重要性 B
- 7. 产褥期护理应注意的问题
- 8. 不同种类的产褥期护理
- 9. 产褥期的营养调配 A
- 10. 产褥期的营养调配 B
- 11. 产后恢复及产后调养 A
- 12. 产后恢复及产后调养 B
- 13. 母乳喂养 A
- 14. 母乳喂养 B
- 15. 母乳喂养 C
- 16. 母乳喂养的好处
- 17. 哺乳指导 A
- 18. 哺乳指导 B
- 19. 母乳喂养的常见问题 A
- 20. 母乳喂养的常见问题 B
- 21. 母乳喂养的常见问题 C
- 22. 产妇护理实践操作

2.7 Fundamental Statistical Genomics

2.8 Machine Learning for Neuropsychiatric Disorders

3 Computer Science

3.1 Deep Learning

1. (a) [Introduction](#)
(b) [Overview of CV on Image Classification](#)
2. [Neural Network](#)
3. [CNN](#)
4. (a) [Training a Neural Network](#)
(b) [Better Accuracy](#)
(c) [Optimizer](#)
(d) [Regularization](#)
(e) [Nuts and Bots of Deep Learning](#)
5. [Advanced CNN Architecture](#)
6. (a) [RNN](#)
(b) [Transformer](#)
- (c) [Vision Transformer](#)
7. (a) [Pruning](#)
(b) [Quantization](#)
(c) [Tensor decomposition](#)
(d) [Low complexity model](#)
8. (a) [Semantic Segmentation](#)
(b) [Object Detection](#)
(c) [Object Detection New Trends](#)
9. (a) [GAN](#)
(b) [Diffusion Model](#)
10. (a) [Pretrained Model and LLM](#)
(b) [Self Supervised Learning](#)

3.2 Causal Data Science

1. [Introduction](#)
2. [Markov Equivalence Class and D-Separation](#)
3. [Markov Equivalence Class and Ancestral Graphs](#)
4. [Partial Ancestral Graph and Fast Causal Inference Algorithm](#)
5. [Conditional Independence Tests & Possible D-Separating Sets](#)
6. [Midterm Review](#)
7. [Effect Identification](#)
8. [Adjustment Criterion for PAGs](#)
9. [Do Calculus](#)
10. [ID Algorithm](#)
11. [ID Algorithm for PAGs](#)

12. Counterfactuals and Effect Estimation

3.3 Concise Machine Learning

1. Introduction; Classification, Data Splitting, and Overfitting versus Underfitting
2. Linear Classifiers, Decision Functions, Boundaries, Centroid Method, and Perceptrons
3. Gradient Descent, Stochastic Gradient Descent, Perceptron Learning, Feature vs Weight Space, Hard-Margin SVM
4. Soft-Margin SVM, Features, and Nonlinear Decision Boundaries
5. Machine Learning Abstractions: Application, Data, Model, Optimization Problem, and Algorithm
6. Decision Theory, Bayes Rule and Risk, Generative and Discriminative Models
7. Gaussian Discriminant Analysis, QDA, LDA, Maximum Likelihood Estimation, Isotropic Gaussian Fitting
8. Eigenvalues, Eigenvectors, Eigendecomposition, Quadratic Forms, Ellipsoids, Covariance, Multivariate Normal Distributions
9. MLE, QDA, and LDA Revisited for Anisotropic Gaussian Distributions
10. Regression, Least Squares, Design Matrix, Normal Equations, Pseudoinverse, Logistic Regression, Gradient Descent
11. Newton's Method, Logistic Regression, LDA Comparison, ROC Curves, Weighted and Polynomial Regression
12. Regression Justification, Empirical Risk, Maximum Likelihood, Bias-Variance Tradeoff, Underfitting and Overfitting
13. Ridge Regression, MAP Regularization, Subset Selection, and Lasso for Reduced Overfitting
14. Decision Trees, Tree-Building Algorithms, Entropy, and Information Gain
15. Decision Tree Regression, Pruning, Multivariate Splits, Ensemble Learning, Bagging, and Random Forests
16. Neural Networks, Gradient Descent, and Backpropagation Algorithm
17. Vanishing Gradients, ReLUs, Softmax Backpropagation, Cross-Entropy Loss, and Computational Model Differences
18. Neuron Biology, Axons, Dendrites, Synapses, Action Potentials, and Faster Training Heuristics

19. Convolutional Neural Networks, Retinal Ganglion Cells, and Simple and Complex Visual Cortex Cells
 20. Unsupervised Learning, PCA, Maximum Likelihood, Variance Maximization, Projection Error Minimization, Eigenfaces
 21. SVD for PCA, K-Means, Hierarchical Clustering, and Greedy Agglomerative Clustering
 22. Pseudoinverse, SVD, Neural Network Generalization, Data Augmentation, L2 Regularization, Dropout, Double Descent
 23. Batch Normalization, ResNets, and Accelerated Optimization with AdamW
 24. AdaBoost, Nearest Neighbor Classification, and Its Relation to Bayes Risk
 25. K-Nearest Neighbor Search, Query Acceleration, Voronoi Diagrams, Point Location, and K-D Trees
- 3.4 Mathematics of Deep Learning
1. The Curse of Dimensionality in Machine Learning
 2. Function Approximation. Linear vs Non-Linear Approximation
 3. Universal Approximation Theorem for Shallow Nets. Harmonic Analysis for Shallow Nets
 4. Reproducing Kernel Hilbert Spaces
 5. Learning in the Overparametrised Regime. Generalisation Bounds for Normed spaces
 6. Convex Optimization Reminders
 7. Optimization in Shallow Spaces: the linear regime. The Neural Tangent Kernel
 8. Nonlinear overparametrised regime. Mean-Field description
 9. Lower bounds: Statistical-Queries, Reductions
 10. Depth Separation
 11. Invariances and Symmetries
 12. Implicit Regularisation. Noisy dynamics. Gradient Flow vs Gradient Descent vs Stochastic Gradient Descent
 13. Generative Modeling: Diffusion

3.5 Generative Artificial Intelligence

1. [Introduction to Machine Learning](#)
2. [Back Propagation and Training Process](#)
3. [Back Propagation and Cost Function](#)
4. [Loss Function of Generative AI](#)
5. [Generative AI Architecture & Entropy Loss Function](#)
6. [Overall Perspectives of Generative AI Architecture](#)
7. [Sequential generative AI with RNN](#)
8. [Back Propagations in RNN](#)
9. [Attention is All You Need](#)
10. [Multi-Head Attentions in Transformer Encoder](#)
11. [Cross-Attentions in Transformer Encoder-Decoder](#)
12. [Generative AI for Image, Generative Adversarial Network Architecture & Loss Functions](#)
13. [Variational Autoencoder Architecture](#)
14. [Variational Autoencoder Loss Function and Image Generation](#)
15. [Diffusion Model for Image Generation](#)
16. [Reinforcement Learning for Fine Tuning and Human Feedback](#)
17. [Markov Property and Bellman Equation](#)
18. [Monte Carlo Method & Temporal Difference Method](#)
19. [Deep Reinforcement Learning for Value Agent Network](#)
20. [Deep Reinforcement Learning for Policy Agent Network](#)

3.6 Nonconvex Optimization for Deep Learning

1. [Linear Regression and GD on Quadratics](#) [A](#) [B](#)
2. [Neural Networks Basics, Stationary Points and Noisy Gradients](#) [A](#) [B](#) [C](#)
3. [Saddle Points and Initialization of Deep MLPs](#) [A](#) [B](#)

4. Non-convex Convergence Rates [A](#) [B](#) [C](#)
 5. SGD Complexity and Learning Rate Scheduling [A](#) [B](#) [C](#)
 6. Generalization and Local Minima [A](#) [B](#) [C](#)
 7. Neural Tangent Kernel [A](#) [B](#) [C](#)
 8. μ P, Hyperparameter Transfer and Overparametrization [A](#) [B](#) [C](#)
 9. Training Deep Networks: Skip Connections and Batch Normalization [A](#) [B](#) [C](#)
 10. Transformers, Layer Norm and Rank Collapse [A](#) [B](#) [C](#)
 11. Adaptive Optimization Methods: Adam and Friends [A](#) [B](#) [C](#)
 12. More Adam and Training LLMs [A](#) [B](#)
 13. Bonus [A](#)
-
1. [Introduction and Course Overview & Convex Sets and Convex Functions](#)
 2. [Convex Functions Continued & Gradient Descent and Family](#)
 3. [Gradient Descent Continued & Projected and Proximal Gradient Descent](#)
 4. [Projected Gradient Descent and Conditional Gradient Descent & Newton, Quasi Newton and Second Order Methods](#)
 5. [Second Order Methods Cont., Barzella Borwein GD and Conjugate Gradient Method & Coordinate Descent Algorithms](#)
 6. [Stochastic Gradient Descent and Family & Variants of SGD for Deep Learning](#) [A](#) [B](#)
 7. [Submodular Functions: Definitions, Examples, and Information Measures](#) [A](#) [B](#)
 8. [Submodular Functions: Definitions, Examples, and Information Measures](#)
 9. [Submodular Function Optimization](#)
 10. [Submodular Function Optimization](#)
 11. [Applications of Continuous and Discrete Optimization](#)

3.7 Machine Learning and Reinforcement Learning

1. Introduction to Machine Learning and Reinforcement Learning
2. Perceptron and Forward Propagation
3. Back Propagation [A](#) [B](#)
4. Entropy and Cost Function [A](#) [B](#)
5. Softmax and Regressions
6. Machine Learning Design and Performance
7. Overview of Deep Reinforcement Learning and Alphago
8. Recurrent Neural Network Architecture
9. Recurrent Neural Network and Long Short-Term Memory (LSTM)
10. RNN Back Propagation Through Time
11. LSTM Back propagation and Transformer Learning
12. CNN Architectures
13. Autoencoder and GAN
14. Markov Decision Process and Bellman Equation
15. Model based Prediction and Control
16. Monte Carlo Method and Temporal Difference Method
17. Value Based DNN Agent
18. Policy Based DNN Agent

3.8 Artificial Intelligence in the Sciences and Engineering

- | | |
|-------------------------------------|--------------------------|
| 1. Course Introduction | 4. Operator Learning |
| 2. Introduction to Deep Learning | (a) Introduction |
| 3. Physics-Informed Neural Networks | (b) FNO |
| (a) Introduction | (c) ReNO |
| (b) Theoretical Insights | (d) Operator Transformer |
| | (e) Graph-based Model |

- (f) [GAOT](#)
- 5. Generative model for PDE: [GenCFD](#)
- 6. Foundation models of PDEs: [Poseidont](#)
- 7. Applications of AI in Chemistry Biology: [A](#) [B](#)

3.9 Biologically Inspired Artificial Intelligence and Optimization

1.
 - (a) [Introduction to the Course & Its Policies & Its Motivation](#)
 - (b) [Evolutionary Approach to Engineering Design Optimization](#)
 - (c) [Population Genetics of Evolutionary Algorithms](#)
 - (d) [The Four Forces of Evolution and The Drift Barrier](#)
 - (e) [The Basic Genetic Algorithm and Its Implementation](#)
 - (f) [GA Wrap: Options for Selection & Crossover & Mutation & Extensions](#)
2.
 - (a) [Evolutionary Computing from Optimization to Programming](#)
 - (b) [Genetic Programming, Immunocomputing & Artificial Immune Systems](#)
 - (c) [Immunocomputing: Genetic Approaches for Diverse Solution Portfolio](#)
3.
 - (a) [Multi-Criteria Decision Making, Pareto Optimality & Intro to MOEA](#)
 - (b) [Multi-Objective Genetic Algorithms: Weight/Vector-Based Approaches](#)
 - (c) [Pareto Ranking and Moving from Communities to Meta-Populations](#)
4.
 - (a) [From DGA/PGA to Niching Methods for Multi-Modal Optimization](#)
 - (b) [Niching Methods in Multi-Modal Optimization](#)
5.
 - (a) [Introduction to Simulated Annealing and Entropy](#)
 - (b) [From Maximum Entropy Toward Optimization by Sim Annealing](#)
 - (c) [Toward SA: Intro to Boltzmann Sampling and Monte Carlo Integration](#)
 - (d) [Simulated Anneal & Wrap and Introduction to Ant Colony Opt](#)
6.
 - (a) [Ant Colony Opt. and Intro. to Bacterial Foraging Opt.](#)
 - (b) [Bacterial Foraging Opt. & Particle Swarm Optimization](#)
7.
 - (a) [Neural Foundations of Learning](#)
 - (b) [Feeding Forward from Neurons to Networks](#)
 - (c) [Recurrent Networks and Temporal Supervision](#)
 - (d) [Reinforcement Learning: Active Learning in Rewarding Environments](#)
 - (e) [Learning without Teacher: Unsupervised & Self-Supervised Learning](#)
 - (f) [Spiking Neural Networks and Neuromorphic Computation](#)
8.
 - (a) [Complex Systems Models of Computation: Cell. Automata & Neighbors](#)

3.10 High Dimensional Analysis: Random Matrices and Machine Learning

1. Introduction and Survey
2. Volumes in High Dimensions
3. Concentration of Volumes
4. Gaussian Random Vectors and Concentration of Their Norm
5. Exponential Concentration of Norm of Gaussian Random Vectors
6. Non-Linear Concentration of Gaussian Random Vectors for Lipschitz Functions
7. Proof of Non-Linear Concentration for Gaussian Random Vectors
8. General Remarks on Linear and Non-Linear Concentration Inequalities
9. Wishart Random Matrices and Concentration of Largest Eigenvalue
10. Proof of Concentration of Largest Eigenvalue
11. The Marchenko-Pastur Law for Wishart Matrices
12. Preparations for Proof of Marchenko-Pastur Law
13. Proof of Marchenko-Pastur: Equation for Stieltjes Transform
14. Proof of Marchenko-Pastur: Stieltjes Inversion Formula
15. Spiked Signal-Plus-Noise Model
16. Proof of the Signal-Plus-Noise Theorem
17. Double Descent and Linear Regression: Over-Determined Case
18. Double Descent and Linear Regression: Under-Determined Case
19. General Remarks on Random Feature Model
20. Asymptotic Eigenvalue Distribution in the Random Feature Model
21. Another Proof of Marchenko-Pastur – Via Stein’s Identity
22. Resolvent Method and Cumulant Expansion
23. Cumulants and Their Properties and Uses
24. Calculation of the Random Feature Eigenvalue Distribution

25. [Gaussian Equivalence Principle for Non-Linear Random Features](#)
 26. [Gradient Descent for Linear Regression](#)
 27. [Time Evolution of Learning and Neural Tangent Kernel](#)
 28. [Properties of the Neural Tangent Kernel](#)
 29. [Free Probability Theory and Linearization of Non-Linear Problems](#)
 30. [The Curse and the Blessing of High Dimensions](#)
- 3.11 [Mathematics of Massive Data Analysis: Fundamentals and Applications](#)
1. [Concentration Inequalities; Geometry of High-Dimensional Euclidean Space](#)
 2. [Hyperspheres and Hyperballs are Weird in High Dimensions](#)
 3. [Gaussian Annulus Theorem and Johnson-Lindenstrauss Lemma](#)
 4. [Markov Chains and MCMC](#)
 5. [Convergence of Random Walks on Undirected Graphs Using Conductance](#)
 6. [More Markov Chains](#)
 7. (a) [Volume via MCMC and PageRank](#)
(b) [Overview](#)
 8. [Overview of Hashing, Sketching, and Random Graph Theory](#)
 9. (a) [Overview of Wavelets, Low-Dimensional Embeddings, and Persistent Homology](#)
(b) [Hashing Theory](#)
 10. [K-Independent Hash Functions](#)
 11. [Frequency Moments of Data Streams and an Idealized Count-Distinct Sketch](#)
 12. [Non-Idealized Count-Distinct Sketch and F2 Sketch](#)
 13. [Frequent Items Sketch, MinHash Sketch](#)
 14. [Erdos-Renyi Random Graphs and Phase Transitions](#)
 15. [Erdos-Renyi Diameter and Isolated Vertices Phase Transitions](#)
 16. [Erdos-Renyi Component Size Phase Transitions](#)

17. [Percolation Theory](#)
18. [Finishing up Percolation Theory, and Intro to Wavelets](#)
19. [Finishing up Haar Wavelet](#)
20. [Necessary Conditions for General Systems of Wavelets](#)
21. [Sufficient Conditions for General Systems of Wavelets](#). The proof of lemma 11.6 is incorrect and revised as computation 11.6.
22. [The Fast Fourier Transform](#). Symmetry if real proof error; complex conjugates added.
23. [Matrix Sketching](#)
24. [VC-Dimension](#)
25. [Generalization Error in Classification](#)
26. [Representing Topological Spaces by Decomposing into Complexes](#)
27. [More Complexes and Introduction to Chains](#)
28. [Computing Simplicial Homology](#)
29. [Morse Functions](#)
30. [Morse-Smale Complex and Piece-Wise Linear Morse Functions](#)
31. [HyperMinHash](#)
32. [Secure Federated Count-Queries](#)

4 Mathematical Foundations

4.1 Analysis

Analysis I

1. [Real Numbers and the completeness](#)
2. [Limits and continuity in metric space](#)
3. [Open and closed sets in metric space](#)
4. [Topological space & The closure of the set](#)
5. [Equivalence of continuous functions & Inheritance Principle & Homeomorphism](#)

6. Product metric spaces & Completeness & Compactness (sequentially compact)
7. Compactness (2) & Bolzano-Weierstrass Thm & Nested seq of nonempty compact sets
8. Continuous functions and Homeomorphisms on compact sets & Connectedness
9. Some properties in topological spaces
10. Accumulation points and isolated points & Dense & Hausdorff space
11. Properties of Connectedness
12. Path connected & Compactness in metric space, Hausdorff space, Topological space(1)
13. Compactness (2) & Sequentially compact implies compact(open cover) (1)
14. Sequentially compact implies compact(open cover) (2) & Totally bounded (1)
15. Totally bounded (2) & Perfect & Cantor set
16. Cantor surjection Theorem
17. Peano Curves & Riemann integrable & Darboux integrable(1)
18. Darboux integrable(2) & Riemann integrable iff Darboux integrable(1)
19. Riemann integrable iff Darboux integrable(2) & Some integrable functions (1)
20. Some integrable functions (2) & Lebesgue number & Zero set
21. Oscillation & Riemann-Lebesgue Theorem
22. Some applications of R-L theorem
23. Fundamental Theorem of Calculus & Convergence in functional space
24. Complete functional space & Convergence function series & Integrals wst unif conv.
25. Derivatives w.s.t uniform convergence & Equicontinuity and Arzela-Ascoli Theorem
26. Heine-Borel Theorem in a function space & Stone-Weierstrass Theorem
27. Fixed point, contraction and ODE
28. Nowhere differentiable continuous function & Baire's Theorem
29. Corollary of Baire's Theorem
30. Multiple integral and Fubini's Theorem

31. [Change of variables formula \(1\)](#)
 32. [Change of variables formula \(2\)](#)
 33. [Banach spaces](#)
 34. [Banach algebra](#)
 35. [Differentiation in Banach Spaces](#)
 36. [Some examples & Integration in Banach spaces \(1\)](#)
 37. [Integration in Banach spaces \(2\)](#)
 38. [Inverse function theorem](#)
 39. [Lebesgue Outer measure](#)
 40. [Lebesgue measurable set & Abstract outer measure](#)
 41. [Measurable Sets form Sigma-Algebra & Measure continuity](#)
 42. [Regularity \(1\)](#)
 43. [Regularity \(2\)](#)
 44. [Every open set is a countable disjoint union of balls plus a zero set](#)
 45. [Measure Product & Inner measure \(1\)](#)
 46. [Inner measure \(2\) & Slice Measure \(1\)](#)
 47. [Slice Measure \(2\) & Lebesgue integrable functions & Monotone Convergence Theorem](#)
 48. [Completed undergraph & Dominated Convergence Thm and Fatou's Lem & Basic properties](#)
 49. [Some basic properties of measurable functions \(2\)](#)
 50. [Cavalieri's principle & Equivalence measurable function definition & Tonelli Thm](#)
 51. [Vitali covering Lemma](#)
 52. [Density Theorem](#)
- Analysis II
1. [Lebesgue Differentiation Theorem \(1\)](#)
 2. [Lebesgue Differentiation Theorem \(2\) & Absolutely continuous functions](#)

3. Indefinite integral is abs conti. & A monotone function has a derivative a.e. (1)
4. A monotone function has a derivative almost everywhere (2)
5. Riemann integrals vs Lebesgue integrals
6. Bounded variation
7. Equi relation of B-V functions and abs conti functions & R-S integrals (1)
8. Riemann-Stieltjes integral (2)
9. Riemann-Stieltjes integral (3)
10. Riemann-Stieltjes integral (4)
11. $L^2 [a, b]$ space
12. Properties of $L^2 [a, b]$ space
13. Orthogonal basis in L^2
14. Fourier coefficients and Parseval's formula
15. A necessary condition of Fourier series converges in L^1 & Riemann-Lebesgue lemma
16. Dirichlet kernel
17. Dini Theorem & There is a continuous function whose Fourier series diverges (1)
18. There is a continuous function whose Fourier series diverges (2) & U-B Principle
19. There is a continuous function whose Fourier series diverges (3)
20. Some properties of Fourier coefficients & Cesàro summability
21. Fejér's Theorem & Approximate integrable functions by tri-polynomial in L^1 norm
22. The average of Fourier series converges in L^p norm
23. Equi. of the average of Fourier series converges unif & Weakly convergence(1)
24. Weakly convergent(2) & Equi condition for tri-series to be the Fourier series
25. Convergence of Cesàro mean of Fourier series (1)
26. Part2 Conv of Cesàro mean of Fourier series(2), Conjugate Fourier series & Dini theorem
27. Singular integrals and conjugate function

28. Bound of the L^2 norm of conjugate function & Hardy-Littlewood Maximal function
29. Bound of the norm of the supremum of conjugate functions
30. A counterexample & C-Z Decomposition & The conjugate function exists a.e. in L^1
31. The conjugate function exists a.e. in L^1 (2)
32. The conjugate function exists a.e. in L^1 (3) & Weakly estimate
33. Applications of Fourier series -isoperimetric inequality
34. Applications of Fourier series -continuous but nowhere differentiable function
35. Fourier transform and some properties
36. Fourier inversion formula and Minkowski's inequality
37. Poisson summation formula & Roth's theorem on 3-term AP
38. Discrete Fourier series, Fourier transform and some properties (1)
39. Discrete Fourier series, Fourier transform and some properties (2)
40. Bourgain's theorem
41. Proof of Roth's theorem on 3-terms AP (step 1)
42. Proof of Roth's theorem on 3-terms AP (step 2 to step 7)
43. Proof of Roth's theorem on 3-terms AP (step 9 , step 8)
44. Proof of Roth's theorem on 3-terms AP (step 8)
45. There is f in L^1 such that for a.e. x , the Fourier series diverges (1)
46. There is f in L^1 such that for a.e. x , the Fourier series diverges (2)
47. There is f in L^1 such that for a.e. x , the Fourier series diverges (3)
48. There is f in L^1 such that for a.e. x , the Fourier series diverges (4)

4.2 Calculus

Calculus I

1. System of real number 1
2. System of real number 2
3. System of real number 3
4. Completeness of Real Number, Weierstrass Theorem 1
5. Weierstrass Theorem 2, define subsequence, Cauchy Sequence
6. Least Upper Bound & Greatest Lower Bound
7. Continuous function, uniformly continuous, Lipchitz continuous
8. Intermediate value theorem, Extreme Value Theory
9. Rieman sum
10. Properties of integral, MVT/general MVT for integral, logarithm
11. Logarithm and exponential 1
12. Logarithm and exponential 2
13. Fundamental Theorem of Calculus 1, Rolle's Theorem, MVT for derivative 1
14. MVT for derivative 2, approximation by linear function, FTC 2
15. Properties of derivative, composite functions, chain rule
16. First/second derivative test
17. Order of magnitude, introduce an smooth function
18. Some example of oscillating functions, chain rule, method of substitution
19. Integration by parts
20. Integration of rational functions 1
21. Integration of rational functions 2, Improper integral 1
22. Improper integral 2
23. Taylor's series: Gamma function, power series, expansion of the logarithm

24. Taylor's series: Taylor's theorem, Cauchy's & Lagrange's remainder
25. Taylor's series: estimate remainder term, expansion of elementary
26. Taylor's series: examples
27. Taylor's polynomial: interpolation 1
28. Taylor's polynomial: interpolation 2, approximation in first order
29. Taylor's polynomial: approximation in second order
30. Taylor's polynomial: fixed point approximation, Stirling's formula
31. Series: concepts of convergence and divergence 1
32. Series: rearrangement, tests for absolute convergence and divergence 1
33. Series: tests for abs. convergence and divergence 2, sequences of functions
34. Series: pointwise/uniform convergence 1
35. Series: pointwise/uniform convergence 2
36. Series: power series, interval of convergence
37. Series: product of two power series
38. Series: expansion of given power series, infinite product
39. Fourier series: periodic function, complex notation 1
40. Fourier series: complex form 2, trigonometric formula, Riemann Lebesgue Lemma
41. Fourier series: examples of Fourier expansion
42. Fourier series: main theorem on Fourier expansion
43. Fourier series: examples of Fourier series 1
44. Fourier series: examples of Fourier series 2, Bessel's Inequality
45. Approximation by trigonometric and rational polynomial 1
46. Approximation by trigonometric and rational polynomial 2
47. Approximation by trigonometric and rational polynomial 3
48. Inner product

- 49. Bernoulli polynomial and their applications 1
- 50. Bernoulli polynomial and their applications 2
- Calculus II
 - 1. Functions of multiple variables & partial derivative
 - 2. Continuity
 - 3. Differentiability & directional derivative (1)
 - 4. Directional derivative (2) & tangent plane
 - 5. Change of variables & Taylor series
 - 6. MVT & Taylor expansion & Integral
 - 7. Double integral & length of curve
 - 8. Curvature & linear differential one form
 - 9. Line integral
- 10. Heine-Borel theorem
- 11. Compact subset & Implicit Function theorem (1)
- 12. Implicit Function theorem (2)
- 13. Inverse Function theorem (1)
- 14. Inverse Function theorem (2) & extreme value
- 15. Lagrange Multiplier method
- 16. Examples & Hölder Inequality
- 17. sufficient condition for local max/min
- 18. Jordan measurable
- 19. Jordan area (1)
- 20. Jordan area (2)
- 21. Double integral (1)
- 22. Double integral (2)

23. Transformation of multiple integrals
24. Improper multiple integrals (1)
25. Improper multiple integrals (2) & volumes
26. Surface area
27. Surface area formula
28. Multiple integral in curved coordinate
29. Extend to $\mathbb{R}^n$
30. Integral of unbounded set
31. Fourier integral theorem (1)
32. Fourier integral theorem (2)
33. Fourier transform - decay property & Parseval's identity
34. Fourier transform of several variables
35. Green theorem
36. Divergence theorem
37. Prove Inverse Function theorem by Green theorem
38. Orientation of surface in $\mathbb{R}^3$ I
39. Orientation of surface in $\mathbb{R}^3$ II
40. Gauss theorem in $\mathbb{R}^3$ I
41. Gauss theorem in $\mathbb{R}^3$ II
42. Application of fluid
43. Stoke's theorem
44. General surface (partition of unit) (1)
45. General surface (partition of unit) (2)
46. General surface (partition of unit) (3) & functions of one complex variable
47. Complex function - power series

48. Complex function - differentiability
49. Conformal map & integration of analytic functions - Cauchy's theorem
50. Complex integral theorem & Cauchy integral formula
51. Zeros, poles and residues

Orthogonal Curvilinear Coordinates

4.3 Lie Algebra

1. Introduction to Lie Groups with Focus on Matrix Groups A
2. Introduction to Lie Groups with Focus on Matrix Groups B
3. Introduction to Lie Groups with Focus on Matrix Groups C
4. Definition, Examples, Lie Subalgebra, Ideals, Adjoint Homomorphism
5. Adjoint Homomorphism, Structure Constants, Cross Product Lie Algebra
6. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical A
7. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical B
8. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical C
9. Solvable, Nilpotent and Simple Lie Algebras A
10. Solvable, Nilpotent and Simple Lie Algebras B
11. Lemmas, Engel and Lie's Theorems, Lie Algebra Homomorphism, Modules A
12. Lemmas, Engel and Lie's Theorems, Lie Algebra Homomorphism, Modules B
13. Representations, Factor Modules, Irreducible Submodules A
14. Representations, Factor Modules, Irreducible Submodules B
15. Schur's Lemma, $SL(2, \mathbb{C})$ Representation on V_d Begins
16. $SL(2, \mathbb{C})$ Representation on V_d Continued, Irreducibility Confirmed

4.4 Linear Algebra

Linear Algebra I

1. Fields, Vector Spaces, Subspaces.
2. Subspaces, Linear Dependence, Basis.
3. Linear Dependence, Basis.
4. Basis, Replacement Theorem, Dimension.
5. Linear Transformation, Kernel, Range, Dimension Theorem.
6. Dimension Theorem, Projection.
7. Matrix Representations.
8. Matrix Representations, Invertible Linear Transformations.
9. Invertible Linear Transformations, Change of Coordinates.
10. Elementary Matrices.
11. Elementary Matrices, Gaussian Elimination.
12. Gaussian Elimination, Determinant of Order 2.
13. Determinants.
14. Determinants(in Terms of Permutations),Cramer's Rule,Adjoint Matrices.
15. Diagonalization
16. Diagonalization
17. Invariant Subspaces, Cayley-Hamilton Theorem
18. Jordan forms, generalized eigenspaces (w. notes)
19. Jordan forms
20. Jordan forms.
21. Jordan forms.
22. Exponential of matrices
23. Systems of first order differential equations, minimal polynomials.

24. Rational canonical forms.

25. Rational canonical forms.

26. Rational canonical forms.

Linear Algebra II

1. Dual Spaces
2. Quotient Spaces
3. Inner Product Space
4. Inner Product Space, Gram-Schmidt Process
5. Orthogonal Projection
6. Hilbert Space, Riesz Representation Theorem
7. Riesz Representation Theorem
8. Adjoint Operators, Normal Operators
9. Normal and Self adjoint Operators
10. Positive semidefinite, Orthogonal and Unitary operators
11. Orthogonal and Unitary operators
12. Orthogonal projection, Spectral theorem
13. Spectral theorem, Simultaneous diagonalization of normal operators
14. Singular value decomposition, Polar decomposition
15. Pseudoinverse
16. Pairs of commuting matrices over a finite field
17. Pairs of commuting matrices over a finite field
18. Bilinear forms
19. Bilinear forms, Quadratic forms
20. Quadratic forms
21. Quadratic spaces

- 22. Quadratic spaces
- 23. Witt decomposition theorem
- 24. Cartan-Dieudonné theorem
- 25. Cartan-Dieudonné theorem
- 26. Bézout's theorem
- 27. Bézout's theorem

4.5 Basic Geometry

4.6 Measure Theory

- 1. Introduction: a Non-Measurable Set
- 2. Classes of Subsets (Semi-Algebras, Algebras and Sigma-Algebras), and Set Functions
- 3. Set Functions
- 4. Caratheodory Theorem
- 5. Monotone Classes
- 6. The Lebesgue Measure I
- 7. The Lebesgue Measure II
- 8. Complete Measures
- 9. Approximation Theorems
- 10. Integration: Measurable and Simple Functions
- 11. Measurable Functions
- 12. Definition of the Integral
- 13. Integral of Simple Functions
- 14. Properties of the Integral I
- 15. Properties of the Integral II
- 16. Theorems on the Convergence of Integrals

17. [Product Measures](#)
 18. [Measure on a Countable Product of Spaces](#)
 19. [Fubini's Theorem](#)
 20. [Hahn-Jordan Theorem](#)
 21. [Radon-Nikodym Theorem](#)
 22. [Almost Sure and Almost Uniform](#)
 23. [Convergence in Measure](#)
 24. [Hölder and Minkowski Inequalities](#)
 25. [L_p Spaces](#)
 26. [From Convergence in Measure to Convergence in L_p](#)
 27. [Bounded Linear Operators in L_p](#)
 28. [Vitali's Covering Lemma](#)
 29. [Differentiability of Functions of Bounded Variations](#)
 30. [Absolutely Continuous Functions](#)
 31. [Decomposition of Distribution](#)
 32. [Cantor Ternary Set and Function](#)
 33. [Solution of Exercise 3.2.4](#)
 34. [Solution of Exercise 6.4.3](#)
 35. [Solution of Exercise 9.2.3](#)
 36. [Materials](#)
- 4.7 Abstract Algebra
- Abstract Algebra I
1. [Basic notion, Symmetry groups](#)
 2. [Cosets, Normal Subgroups, Quotient Groups, Isomorphism Theorem](#)
 3. [Isomorphism Theorem, Tower of Subgroups](#)

4. Butterfly lemma, Jordan-Hölder theorem
5. Jordan-Hölder theorem, Cyclic groups
6. Group Action, Simplicity of A_n
7. Simplicity of A_n , Sylow theorems
8. Sylow theorems
9. Semidirect products
10. Structure theorem of finite abelian groups
11. Ring Isomorphism Theorem
12. Chinese Remainder Theorem, Localization
13. ED, PID, UFD
14. Gauss lemma, Criteria for irreducibility
15. Algebraic extensions
16. Algebraic closures
17. Algebraic closures, Normal extensions
18. Separable extensions
19. Primitive element theorem, Finite fields
20. Galois theory
21. Galois theory
22. Galois theory, roots of unity, cyclotomic extensions.
23. Norms and traces.
24. Purely inseparable extensions, cyclic extensions.
25. Solvable extensions, solvable by radicals.

Abstract Algebra II

1. Module theory.
2. Free modules, projective modules.

3. Modules over PID.
4. Modules over PID.
5. Tensor products.
6. Modules of fractions.
7. Noetherian modules.
8. Primary decompositions.
9. Primary decompositions, Nakayama's lemma.
10. Nakayama's lemma, filtered and graded modules, Artin-Rees.
11. Hilbert polynomial, Artinian modules.
12. Abelian categories, general homology theory.
13. Injective modules.
14. Homotopies of morphisms of complexes, derived functors.
15. Ext functor.
16. Ext functor, Ext^1 , Tor functor.
17. Group cohomology.
18. Group cohomology, Semisimple modules.
19. Semisimple modules, Semisimple rings.
20. Density theorem, Wedderburn theorem, Burnside theorem.
21. Representations, Characters.
22. Characters, Class functions.
23. Orthogonality relations.
24. Orthogonality relations.
25. Induced characters, Induced representations.
26. Induced representations, Mackey's criterion.
27. Artin theorem, Brauer theorem.

4.8 Complex Analysis

1. 0920 Introduction to the course, examples of applications, definition of holomorphic functions, algebraic stability properties of holomorphic functions.
2. 0921 Convergent power series are holomorphic. Examples and counterexample (the complex conjugate function).
3. 0927 Holomorphy and differentiability; the Cauchy-Riemann equations. Line integrals.
4. 0928 Line integrals and primitives.
5. 1004 Chapter 3: Cauchy's Theorem. Goursat's Theorem, existence of primitives in a circle, Cauchy's Integral Formula.
6. 1005 Chapter 3: proof of Goursat's Theorem.
7. 1011 Chapter 4: applications of Cauchy's Theorem and integral formula: analyticity, Cauchy's inequalities for derivatives, Liouville's Theorem.
8. 1012 Chapter 4: zeros of holomorphic functions, analytic continuation.
9. 1018 Chapter 4: proof of the principle of analytic continuation. Limits of holomorphic functions, Morera's theorem.
10. 1019 Chapter 4: holomorphic functions defined by integrals
11. 1025 Chapter 5: singularities and meromorphic functions, residue theorem.
12. 1026 Chapter 5: residue theorem and examples.
13. 1101 Chapter 5: meromorphic functions, counting zeros, open image and maximum modulus principle.
14. 1102 Chapter 5: meromorphic functions, counting zeros, open image and maximum modulus principle.
15. 1115 Chapter 6: Eta, THeta, Zeta (a long example). Definitions of the functions, infinite products..
16. 1116 Chapter 6: Eta, THeta, Zeta. Analytic continuation of the zeta function, application to prime numbers.
17. 1121 Chapter 6: Eta, THeta, Zeta (a long example). Sketch of Riemann's approach to counting primes; the Riemann Hypothesis.

18. 1123 Chapter 7: Homotopy and applications. Definition and statement of Cauchy's Theorem for homotopic curves.
19. 1129 Chapter 7: Proof of Cauchy's Theorem for homotopic curves.
20. 1130 Chapter 7: simply-connected open sets, existence of primitives. The complex logarithm.
21. 1206 Chapter 7: The residue theorem and homotopy; winding numbers.
22. 1207 Chapter 8: conformal mapping (definition, first examples).
23. 1213 Chapter 8 (conformal mapping): more examples, statement of Riemann's mapping theorem. Outline of the proof. Schwarz Lemma, automorphisms of the disc.
24. 1214 Chapter 8 (conformal mapping); reduction of Riemann's Theorem to the existence of an extremum.
25. 1220 Chapter 8 (conformal mapping): end of the proof of Riemann's Theorem; Montel's Theorem. Final remarks.
26. 1221 Review of the course, questions

4.9 Matrix Lie Groups

- | | |
|---|--|
| 1. Complex and Quaternions A | 10. Lie Groups and Tangent Space at I B |
| 2. Complex and Quaternions B | 11. What is the Logarithm A |
| 3. More Quarternions, Group Theory A | 12. What is the Logarithm B |
| 4. More Quarternions, Group Theory B | 13. On the Derivative of a Homomorphism |
| 5. Chapter 2 and Isometries over R,C and H A | 14. Adjoint and Lie Algebra A |
| 6. Chapter 2 and Isometries over R,C and H B | 15. Adjoint and Lie Algebra B |
| 7. Path Connectedness for $SO(n)$ and $SU(2)$ A | 16. Lie Algebra Example 2nd Order BCH A |
| 8. Path Connectedness for $SO(n)$ and $SU(2)$ B | 17. Lie Algebra Example 2nd Order BCH B |
| 9. Lie Groups and Tangent Space at I A | 18. BCH identity A |
| | 19. BCH identity B |
| | 20. BCH calculation A |
| | 21. BCH calculation B |
| | 22. Universal Covers and Lie's Theorem A |

- 23. Universal Covers and Lie's Theorem B
- 24. Group Orbits and Homogeneous Space A
- 25. Group Orbits and Homogeneous Space B

4.10 Functional Analysis

- 1. Linear Spaces: Definition, Examples and Linear Span.
- 2. Linear Spaces: Quotient Spaces and Convex Sets.
- 3. Normed Linear Spaces: Definition and Basic Properties.
- 4. Completing a Normed Linear Space.
- 5. Finite Dimensional Linear Spaces.
- 6. Examples of Normed Linear Spaces.
- 7. In Infinite Dimensions the Unit Ball is not Compact.
- 8. Zorn's Lemma.
- 9. The Hahn-Banach Theorem.
- 10. Convex Sets and Gauge Functions.
- 11. Geometric Hahn-Banach Theorems.
- 12. Dual of a Normed Linear Space.
- 13. Extension of Bounded Linear Functionals, Closed Linear Spans.
- 14. Reflexive Spaces.
 - (a) The Dual Space of $C([a,b])$.
 - (b) An Application of the Hahn-Banach Theorem: the Moment Problem and Chebyshev Approximation.
 - (c) A Dual Variational Problem in Optimal Control.
 - (d) An Application of the Hahn-Banach Theorem: the Existence of a Green Function.
- 15. Hilbert Spaces.
- 16. Closed Convex Subsets of a Hilbert Space.
- 17. Riesz and Lax-Milgram Representation Theorems.
- 18. Orthonormal Sets and Closed Linear Spans.

19. Orthonormal Bases.
 - (a) A Quadratic Variational Problem.
 - (b) The Dirichlet Principle.
 - (c) Generalized Derivatives and Sobolev Spaces.
20. Uniform Boundedness Principle.
21. Weak Convergence.
22. Uniform Boundedness of Weak Converging Sequences.
23. Weak Sequentially Compactness.
24. Weak* Topology.
25. Applications of Weak Convergence.
26. Bounded Linear Operators.
27. Transpose of Bounded Linear Operators.
28. Strong and Weak Convergence of Operators.
29. Principle of Uniform Boundedness for Maps and Compositions.
30. Open Map Principle.
31. The Closed Graph Theorem.
32. Examples of Bounded Linear Maps: Integral Operators.
33. Symmetric Operators.
34. Eigenvalues of Compact Symmetric Operators.
35. The Fredholm Alternative.
36. An Application to Integral Operators.
37. Materials

4.11 Mathematical Modeling

4.12 Partial Differential Equations

1. A. Intro, definitions, elementary examples.
B. Advection equations. Characteristic lines. General solutions of quasi-linear first order equations.
2. First order quasi-linear PDEs in two variables. Solution surface. Characteristic equations. Cauchy problem. Examples: advection equations and Burgers equation.
3. Burgers equation. Gradient catastrophe, weak solutions, shock formation, properties and propagation.
4. A. Shock waves, weak solutions, speed of propagation in Burgers equation. Conservation laws.
B. Parametric solution to general first order PDEs. Monge cone. Characteristic equations.
5. Cauchy problem for general first order PDEs. Wave fronts on an ellipse, caustics. ODEs general solution for a linear system.
6. Second order PDEs intro. Wave equation D'Alembert solution. Changes of variables. Characteristic coordinates.
7. Linear and quasi-linear second-order equations. Characteristic curves.
8. Linear second-order equations. Example: Tricomi equations. Elliptic equations.
9. Parabolic linear second order equations. Examples. Rescaling the dependent variable to remove first order terms.
10. A. Elliptic equations. Laplace equation solutions and Cauchy's theorem. On a disk.
B. Elliptic equations. Laplace equation solutions and Cauchy's theorem. On the upper half plane.
11. Uniqueness theorems, Laplace equation, divergence theorem, separation of variables, mean value theorem.
12. Maximum minimum principle for elliptic equations. Green's functions. Symmetrical solutions, boundary conditions.
13. Symmetry properties of the Green's function. Boundary conditions of the Green's function: Dirichlet, Robin, Neumann.

14. A. Green's function for Laplace equation on a disk.
B. Green's function for Laplace equation on a rectangle.
 15. Diffusion equation, self-similar solutions, uniqueness, min-max principle.
 16. Green's function for the heat equation. Infinite and semi-infinite domains, Neumann and Dirichlet boundary conditions.
 17. Green's function for the 1D diffusion equation on a finite domain. Wave equation in 2D and 3D, preliminaries. Uniqueness of solutions.
 18. Fourier transform properties. Green's function for the wave equation in nD.
 19. Green's function for the wave equation in 1D and 2D.
 20. Wave equation. Green's function in 3D. Example: Radiation from an electron on an antenna.
 21. Separation of variables in 2D. Laplace, Helmholtz, Schrödinger potentials. Separable coordinate systems in 2D.
 22. Sturm-Liouville oscillation theory, Prüfer substitution, comparison theorems.
 23. A. Sturm-Liouville theorem. Eigenvalue problem. Properties of zeros, dependence on the eigenvalue. Existence of unique orthogonal basis.
B. Expanding functions on a Sturm-Liouville eigenbasis. Properties and method.
 24. Wave equation on a disk, separation of variables.
 25. Wave equation on a circular membrane, Bessel functions, zeros, generating function, integral representations.
 26. Wave equation on a circular membrane, expansions using Bessel functions.
 27. Solitons, Korteweg-de Vries equations. Advection nonlinear velocity, dispersion.
 28. KdV travelling wave solutions, self-similar solutions, inverse scattering method, solitons.
- 4.13 Ordinary Differential Equations
1. Some Basic Mathematical Models
 2. Separable Differential Equations
 3. Linear v.s. Non-Linear Differential Equations
 4. Linear First Order ODE

5. [Applications of Separable Equations](#)
6. [Applications](#)
7. [Existence and Uniqueness, Linear v.s. Nonlinear](#)
8. [Autonomous First Order ODEs, Population Dynamics](#)
9. [Exact Differential Equations](#)
10. [Euler's Method](#)
11. [Review](#)
12. [2nd Order Linear ODEs](#)
13. [2nd Order Linear Homogeneous ODEs with Constant Coefficients](#)
14. [Complex Roots of the Characteristic Equation](#)
15. [Reduction of Order](#)
16. [Nonhomogeneous 2nd Order Linear Equations, Method of Undetermined Coefficients](#)
17. [Variation of Parameters](#)
18. [Vibrations](#)
19. [Higher Order Linear ODEs](#)
20. [Forced Vibrations](#)
21. [Linear Independence](#)
22. [n-th Order Linear ODEs with Constant Coefficients](#)
23. [Variation of Parameters for n-th Order Linear ODEs](#)
24. [Method of Variation of Parameters for n-th Order Linear ODEs](#)
25. [Systems of First-Order Linear Equations](#)
26. [On \$2 \times 2\$ Linear Systems with Constant Coefficients](#)
27. [Linear 1st Order Homogeneous \$2 \times 2\$ Systems with Constant Coefficients](#)
28. [Linear Systems with Constant Coefficients](#)
29. [\$3 \times 3\$ Systems , Fundamental Matrices](#)

30. [Repeated Roots](#)
31. [Nonhomogeneous First Order Linear Systems](#)
32. [Non-linear \$2 \times 2\$ Systems](#)
33. [Competing Species](#)
34. [Review](#)
35. [Analyzing Non-Linear Systems](#)
36. [Damped Pendulum with a Stiff String](#)
37. [Predator Prey Equations](#)
38. [Laplace Transform](#)
39. [Using Laplace Transform](#)
40. [Laplace Transform of Picewise Defined Functions](#)
41. [Review](#)
42. [Review](#)